MILITARY SPECIFICATION

RECTIFIERS, METALLIC, SELENIUM

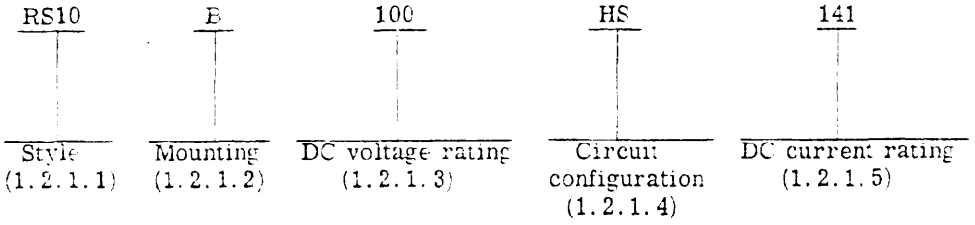
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers selenium metallic rectifiers consisting of one or more cells.

1.2 Classification.

1.2.1 Type designation. The type designation shall be in the following form, and as specified (see 3.1 and 6.2):



1.2.1.1 Style. The style is identified by the two-letter symbol "RS" followed by a two-digit number; the letters identify selenium metallic rectifiers and the number identifies the cell size in accordance with table I.

TABLE I. Style.

Symbol	Cell size	Symbol	Cell size
	Inches		Inches
RS10 - - -	1 x 1	RS50 - - -	3 x 3
RS20 - - -	1.25 x 1.25	RS60 - - -	4 x 4
RS30 - - -	1.6 x 1.6	RS70 - - -	4 x 6
RS40 - - -	2.2 x 2.2	RS80 - - -	5 x 6

1.2.1.2 Mounting. The method of mounting is identified by a single letter in accordance with table II.

TABLE II. Mounting method.

Symbol	Mounting method
B - - - - - -	Bolt
S - - - - - -	Stud

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1.2.1.3 DC voltage rating. The nominal direct-current (dc) output voltage rating, at 55° C ambient temperature, when the rectifier is connected to a resistive load, is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow.

1.2.1.4 Circuit configuration. The circuit configuration is identified by a two-letter symbol in accordance with table III.

TABLE III. Circuit configuration.

Symbol	Circuit configuration
HS - - - - - -	Half wave, single phase
CS - - - - - -	Center tap, single phase
BS - - - - - -	Bridge, single phase
DS - - - - - -	Doubler, single phase
HT - - - - - -	Half wave, three phase
CT - - - - - -	Center tap, three phase
BT - - - - - -	Bridge, three phase

1.2.1.5 DC current rating. The nominal dc output current rating, at 55° C ambient temperature, when the rectifier is connected to a resistive load, is identified by a three-digit number; the first two digits represent significant figures in milliamperes and the last digit specifies the number of zeros to follow.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-S-85 - Screw, Cap, Slotted- and Hexagon-Head.
QQ-C-320 - Chromium Plating (Electrodeposited).
QQ-N-290 - Nickel Plating (Electrodeposited).
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-S-365 - Silver Plating (Electrodeposited), General Requirements For.
QQ-Z-325 - Zinc Coating (Electrodeposited), Requirements For.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Setup.
PPP-T-60 - Tape; Pressure-Sensitive Adhesive, Waterproof for Packaging.
PPP-T-76 - Tape; Pressure-Sensitive Adhesive, Paper (For Carton Sealing).

MILITARY

MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-79 - Plastic Rod and Tube, Thermosetting, Laminated.
MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shock Tests, HI (High Impact), Shipboard Machinery, Equipment and Systems, Requirements For (Navy).
MIL-S-6872 - Soldering Process, General Specification For.

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SPECIFICATIONS - Continued

MILITARY - Continued

MIL-P-15037 - Plastic Sheet, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin.
MIL-R-18281 - Rectifiers, Metallic, Selenium, Copper Oxide and Magnesium-Copper Sulfide Type, Naval Shipboard Use.
MIL-C-45662 - Calibration System Requirements.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-456 - Electronic Parts, Date and Source Coding For.

(See supplement 1 for list of associated specification sheets.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements should be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. Selenium rectifiers furnished under this specification shall be a product which has been tested and has passed the first article tests specified in 4.4 (see 6.4).

3.3 Material. The material for each part shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the rectifiers to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Laminated thermosetting plastic-material. Laminated thermosetting plastic-material shall be type GMG in accordance with MIL-P-79 or MIL-P-15037, as applicable.

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3.3.2 Molded thermosetting plastic-material. Molded thermosetting plastic-material shall be type MAI-60 in accordance with MIL-M-14.

3.3.3 Fungus-resistant materials. As far as practicable, all non-metallic parts shall be fungus resistant. When materials that are not fungus resistant are used, requirement 4 of MIL-STD-454 shall apply.

3.3.4 Screws, bolts, nuts, and threads. Screws, bolts, nuts, and other threaded parts shall be of corrosion-resisting material, or shall be suitably protected against corrosion (see 3.3.5 and 3.3.5.1). All external threaded parts shall be in accordance with Handbook H28.

3.3.5 Corrosion-resisting materials. Silver, corrosion-resisting steel, copper, brass, copper-nickel, and nickel-copper alloys are considered satisfactory corrosion-resisting materials.

3.3.5.1 Corrosion-resisting treatment. Electroplating with cadmium in accordance with QQ-P-416, type II, class B; chromium in accordance with QQ-C-320; nickel in accordance with QQ-N-290, type III; zinc in accordance with QQ-Z-325, class 2, type II; or silver in accordance with QQ-S-365 is considered a satisfactory corrosion-resisting treatment.

3.4 Design and construction. Rectifiers shall be of the design, construction, and physical dimensions specified (see 3.1).

3.4.1 Terminals and connections. Each rectifier shall be completely wired, and terminals shall be located so as to be readily accessible.

3.4.1.1 Terminal binding screws. For rectifiers constructed with cells with an overall area larger than 9 square inches, terminal binding screws shall be provided for all terminals for connections external to the rectifier. The screws shall be brass with two type A nuts in accordance with FF-S-85. The screw size shall be in accordance with table IV.

TABLE IV. Terminal binding screws.

Current rating, maximum (amperes)	Screw size
15 and under	6-32
Over 15 to 25 inclusive	8-32
Over 25 to 50 inclusive	10-32
Over 50 to 100 inclusive	1/4-20

3.4.1.2 Terminals for soldering. Rectifiers constructed with cells with an overall area of 9 square inches or less shall have terminals suitable for soldering. Terminals shall be such that wires can be hooked, wrapped around, or otherwise mechanically secured to the terminals prior to soldering, in such a manner that solder alone will not be depended upon for mechanical strength.

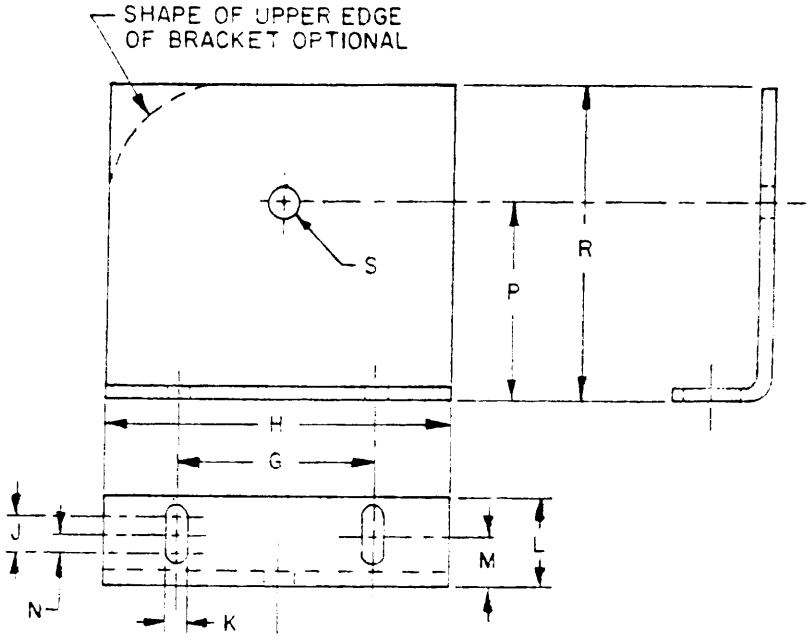
3.4.2 Mounting brackets. Mounting brackets for normal applications are shown on figure 1.

3.4.2.1 High-impact shock. Mounting brackets for high-impact shock application are shown on figure 3 of MIL-R-18281.

3.5 Forward voltage drop. When tested as specified in 4.6.2, rectifiers shall have a forward voltage drop not exceeding the value specified (see 3.1).

3.6 Reverse current. When tested as specified in 4.6.3, rectifiers shall have a reverse current not exceeding the value specified (see 3.1).

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Style	Dimensions ±.02 (.51)							Bracket Mtg screw	S ±.02 (.51)
	G	J	K	L	M	N	P		
RS10	.56 (14.22)	.19 (4.83)	.16 (4.06)	.62 (15.75)	.31 (7.87)	.09 (2.29)	1.06 (26.92)	6-32	.18 (4.57)
RS20	.56 (14.22)	.19 (4.83)	.16 (4.06)	.62 (15.75)	.31 (7.87)	.09 (2.29)	1.19 (30.23)	6-32	.22 (5.59)
RS30	.56 (14.22)	.19 (4.83)	.16 (4.06)	.62 (15.75)	.31 (7.87)	.09 (2.29)	1.38 (35.05)	6-32	.22 (5.59)
RS40	1.12 (28.45)	.25 (6.35)	.22 (5.59)	.88 (22.35)	.50 (12.70)	.12 (3.05)	1.75 (44.45)	10-32	.27 (6.86)
RS50	1.50 (38.10)	.28 (7.11)	.28 (7.11)	1.00 (25.40)	.59 (14.99)	.14 (3.56)	2.19 (55.63)	1/4-20	.27 (6.86)
RS60	1.50 (38.10)	.28 (7.11)	.28 (7.11)	1.00 (25.40)	.59 (14.99)	.14 (3.56)	2.69 (68.33)	1/4-20	.32 (8.13)
RS70	2.50 (63.50)	.28 (7.11)	.28 (7.11)	1.00 (25.40)	.59 (14.99)	.14((3.56)	3.50 (88.90)	1/4-20	.32 (8.13)
RS80	2.50 (63.50)	.28 (7.11)	.28 (7.11)	1.00 (25.40)	.59 (14.99)	.14 (3.56)	4.00 (101.60)	1/4-20	.32 (8.13)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Dimension H shall not exceed the width of the cell; dimension R shall not extend beyond the top of the rectifier cell.
4. Millimeters are in parentheses.

FIGURE 1. Mounting brackets for normal applications.

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3.7 Dielectric strength. When tested as specified in 4.6.4, rectifiers shall withstand the potential specified in 4.6.4 without damage, arcing, or breakdown.

3.8 Insulation resistance. When tested as specified in 4.6.5, rectifiers shall have an insulation resistance of not less than 100 megohms.

3.9 Low-temperature exposure. When rectifiers are tested as specified in 4.6.6, the forward voltage drop shall change not more than 8 percent from the initial value obtained when measured as specified in 4.6.2. There shall be no peeling of the protective finish; cracking of the protective finish without peeling will be acceptable.

3.10 Moisture resistance. When rectifiers are tested as specified in 4.6.7, the forward voltage drop shall change not more than 20 percent from the initial value obtained when measured as specified in 4.6.2, the reverse current shall not exceed two times the value specified (see 3.1), there shall be no dielectric breakdown, and the insulation resistance shall be not less than 20 megohms.

3.11 Low-temperature operation. When rectifiers are tested as specified in 4.6.8, the forward voltage drop shall change not more than 100 percent from the initial value obtained when measured as specified in 4.6.2, and the reverse current shall not exceed two times the value specified (see 3.1).

3.12 High-temperature operation. When rectifiers are tested as specified in 4.6.9, the reverse current shall not exceed two times the value specified (see 3.1).

3.13 Life. When rectifiers are tested as specified in 4.6.10, the forward voltage drop during the first hour of the test shall not exceed the value specified (see 3.1). At the completion of the test, the forward voltage drop shall change not more than 100 percent from the value obtained during the first hour of the test.

3.14 Mechanical shock. When rectifiers are tested as specified in 4.6.11, the forward voltage drop shall change not more than 5 percent from the initial value obtained when measured as specified in 4.6.2, the reverse current shall change not more than 5 percent from the initial value obtained when measured as specified in 4.6.3, and there shall be no dielectric breakdown.

3.15 Vibration. When rectifiers are tested as specified in 4.6.12, the forward voltage drop shall change not more than 5 percent from the initial value obtained when measured as specified in 4.6.2, the reverse current shall change not more than 5 percent from the initial value obtained when measured as specified in 4.6.3, and there shall be no dielectric breakdown. There shall be no looseness of parts or mechanical damage.

3.16 Corrosion. When rectifiers are tested as specified in 4.6.13, the forward voltage drop shall change not more than 8 percent from the initial value obtained when measured as specified in 4.6.2, the reverse current shall change not more than 8 percent from the initial value obtained when measured as specified in 4.6.3, and there shall be no dielectric breakdown. There shall be no peeling of the protective finish; cracking of the protective finish without peeling will be acceptable.

3.17 Marking. Rectifiers shall be marked in accordance with MIL-STD-130 and as specified in 3.17.1 or 3.17.2, as applicable. Terminals shall be color coded in accordance with 3.17.3. Marking shall remain legible, and color coding shall be distinct, after all tests.

3.17.1 Rectifiers with large cells. Rectifiers constructed with cells larger than 2.2 inches square shall be marked with the type designation, manufacturer's name or approved code designation, manufacturer's catalog number or identification symbols for the rectifier, and the date of manufacture. The date of manufacture shall be shown by the date code specified in MIL-STD-456.

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3.17.2 Rectifiers with small cells. Rectifiers constructed with cells 2.2 inches square or less shall be marked with the type designation and the manufacturer's name or approved code designation.

3.17.3 Color coding of terminals. Positive terminals shall be color-coded red; negative terminals shall be color-coded black; common terminals which are both positive and negative shall be color-coded yellow.

3.18 Workmanship. Rectifiers shall be manufactured and processed in a careful and workmanlike manner, in accordance with good design and sound practice.

3.18.1 Soldering. Soldering shall be in accordance with MIL-S-6872. Solder shall be Sn40 or a composition with a higher tin content.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The supplier shall establish and maintain a calibration system in accordance with MIL-C-45662.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.3.1 Cooling. Unless otherwise specified, rectifiers shall be cooled by natural convection.

4.3.2 AC test voltages. AC test voltages shall be 60 Hertz frequency and approximately sinusoidal, having a total harmonic voltage distortion of not more than 7 percent.

4.4 First article inspection. First article inspection shall be performed by the supplier, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract or purchase order under which it is granted, unless extended by the Government to the contracts or purchase orders.

4.4.1 Sample size. Four rectifiers shall be subjected to first article inspection.

4.4.2 Inspection routine. The sample shall be subjected to the inspections in table V, in the order shown.

4.4.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

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ample units are rejected order provided all damage is repaired and the s

TABLE V. First article inspection.

Examination or test	Requirement paragraph	Method paragraph	Number of sample units to be inspected	Number of failures
Visual and mechanical inspection 1/	3.3 to 3.4.2.1 incl, and 3.17 to 3.18.1 incl	4.6.1	4	0
Forward voltage drop	3.5	4.6.2.1		
Reverse current	3.6	4.6.3.1		
Dielectric strength	3.7	4.6.4		
Insulation resistance	3.8	4.6.5		
Low-temperature exposure	3.9	4.6.6		
Moisture resistance	3.10	4.6.7		

1/ Marking will be considered defective only if it is illegible at the completion of any of the required tests.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection. Except as specified in 4.5.1.5.3, delivery of products which have passed the group A and B inspection shall not be delayed pending the results of the group C inspection.

4.5.1.1 Inspection lot. An inspection lot shall consist of all rectifiers of the same style, but shall include items having different voltage and current ratings, and circuit configurations.

4.5.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table VI, in the order shown. One additional sample unit shall be included in each sample of half-wave rectifiers (see 4.6.2 and 4.6.3). The additional sample unit shall be similar in electrical characteristics to the rest of the sample units. For doubler rectifiers, twice the number of sample units required shall be taken from the lot.

4.5.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table VI. Major and minor defects shall be as defined in MIL-STD-105.

4.5.1.3 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE VI. Group A inspection.

Examination or test	Requirement paragraph	Test method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical inspection	3.3 to 3.4.2.1 incl, and 3.17 to 3.18.1 incl	4.6.1	1.0	4.0
Forward voltage drop	3.5	4.6.2.2	1.0	---
Reverse current	3.6	4.6.3.2		
Dielectric strength	3.7	4.6.4		
Insulation resistance	3.8	4.6.5		

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4.5.1.4 Group B inspection. Group B inspection shall consist of those tests specified in table VII, in the order shown. They shall be performed on sample units that have been subjected to and have passed the group A tests, unless the Government considers it more practical to select a separate sample from the lot for group B tests. All sample units shall be selected without regard to their quality.

TABLE VII. Group B inspection.

Examination or test	Requirement paragraph	Test method paragraph
Low-temperature operation - - - - -	3.11	4.6.8
High-temperature operation - - - - -	3.12	4.6.9

4.5.1.4.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection levels. The AQL shall be 6.5 (percent defective), and the special inspection level shall be S-4.

4.5.1.4.2 Disposition of sample units. Sample units subjected to group B tests shall not be delivered on the contract or order.

4.5.1.4.3 Action in case of lot rejection. If a lot is rejected due to failure to pass group B tests, the manufacturer may rework the lot or screen out defectives and resubmit it for Government inspection. Resubmitted lots shall be kept separate from new lots. The resubmitted lot shall be inspected, using tightened inspection.

4.5.1.5 Group C inspection. Group C inspection shall consist of those tests specified in table VIII. Tests shall be performed in the order shown. Different sample units shall be used for the quarterly and the semiannual tests. No defectives will be permitted.

TABLE VIII. Group C inspection.

Examination or test	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of defectives permitted
Quarterly				
Low-temperature exposure - - - - - -	3.9	4.6.6	4	0
Moisture resistance - - - - - - - - - -	3.10	4.6.7		
Life - - - - - - - - - - - - - - - - - -	3.13	4.6.10		
Semiannually				
Mechanical shock - - - - - - - - - - -	3.14	4.6.11	4	0
Vibration- - - - - - - - - - - - - - - -	3.15	4.6.12		
Corrosion - - - - - - - - - - - - - - -	3.16	4.6.13		

4.5.1.5.1 Sampling procedure. A separate sample shall be selected for the quarterly and the semiannual tests. Once each quarter, four basic bridge sample units which have passed group A tests shall be selected for the quarterly tests. These sample units shall be of the type in current production with the largest plate area. Semiannually, eight basic bridge sample units which have passed group A tests shall be selected for the semiannual tests. These sample units shall be either style RS10 or the next largest style in current production.

4.5.1.5.2 Disposition of sample units. Sample units subjected to group C tests shall not be delivered on the contract or order.

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4.5.1.5.3 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the Government). In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the contracting officer or purchaser.

4.5.2 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

4.6 Methods of examination and test.

4.6.1 Visual and mechanical inspection. Rectifiers shall be inspected to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.3 to 3.4.2.1 inclusive, and 3.17 to 3.18.1 inclusive).

4.6.2 Forward voltage drop (see 3.5).

4.6.2.1 Forward voltage drop (short-circuit test). Rectifiers shall be connected as shown on figure 2 or 3, as applicable. The forward voltage drop shall be measured after 5 ±1/2 minutes of operation at rated current (see 3.1). The nominal voltage drop of the ammeter shall be not greater than 50 millivolts, and the voltage drop in the output circuit (output terminals of the rectifiers) shall be not greater than 100 millivolts.

4.6.2.2 Forward voltage drop (for group A acceptance tests only). Rectifiers shall be connected as shown on figure 2 or 3, as applicable. The forward voltage drop shall be measured immediately (cold) after application of rated current (see 3.1). The additional half-wave rectifier sample unit (see 4.5.1.2) shall be utilized as the load balance (see figure 2 and 3). Adequate precaution shall be taken to obtain assurance that the additional sample unit is similar in electrical characteristics to the rest of the sample units. Doubler rectifiers shall be tested in the bridge test circuit shown on figure 2.

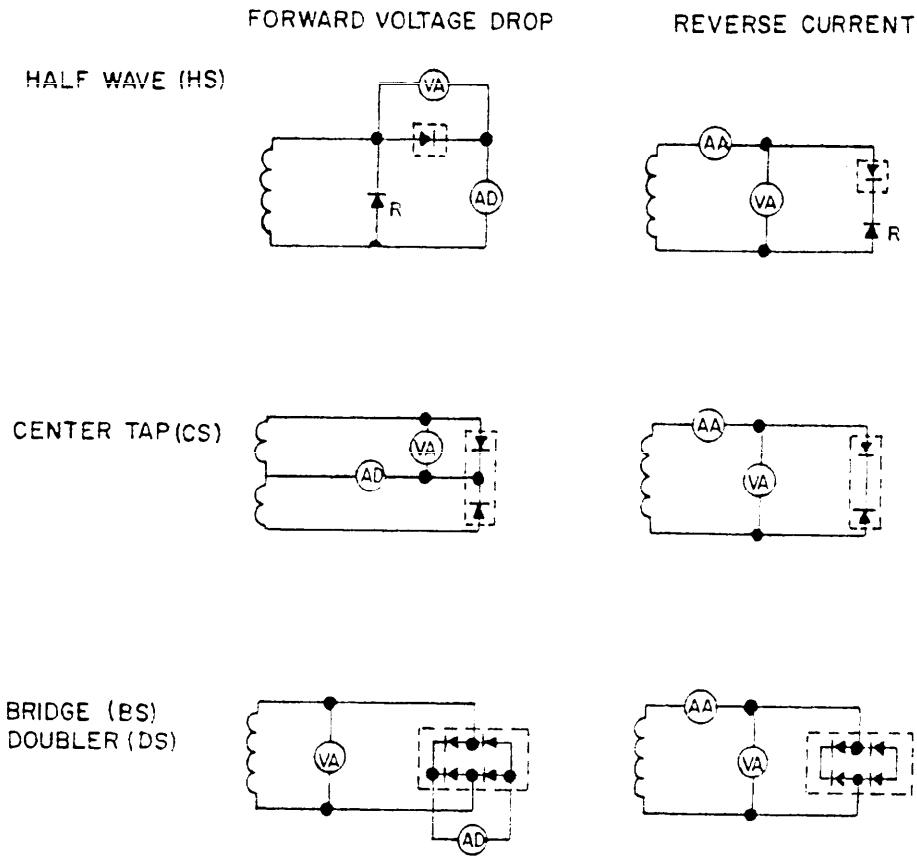
4.6.3 Reverse current (see 3.6).

4.6.3.1 Reverse current (open-circuit test). Rectifiers shall be connected as shown on figure 2 or 3, as applicable. The reverse current shall be measured after 5 ±1/2 minutes of operation at the specified ac applied voltage (see 3.1). The ac applied voltage shall be measured by means of an ac voltmeter of sufficiently high impedance to reduce error due to excessive current drain by the voltmeter, or correction shall be made for the current drain of the voltmeter.

4.6.3.2 Reverse current (for group A acceptance tests only). Rectifiers shall be connected as shown on figure 2 or 3, as applicable. The reverse current shall be measured within 5 minutes after application of the specified ac applied voltage (see 3.1). The additional half-wave rectifier sample unit (see 4.5.1.2) shall be utilized as the load balance (see figure 2 and 3). Adequate precaution shall be taken to obtain assurance that the additional sample unit is similar in electrical characteristics to the rest of the sample units. Doubler rectifiers shall be tested in the bridge test circuit shown on figure 2.

4.6.4 Dielectric strength. Rectifiers shall be subjected to a test potential as specified in table IX for a period of 1 minute for qualification tests and for not less than 5 seconds nor more than 1 minute for acceptance tests. The test potential shall be applied between all current-carrying parts connected together and the mounting members. Voltages above 1,000 volts rms shall be applied gradually at a rate not exceeding 500 volts rms per second; however, for acceptance tests, the test voltage may be applied instantaneously at the option of the manufacturer (see 3.7).

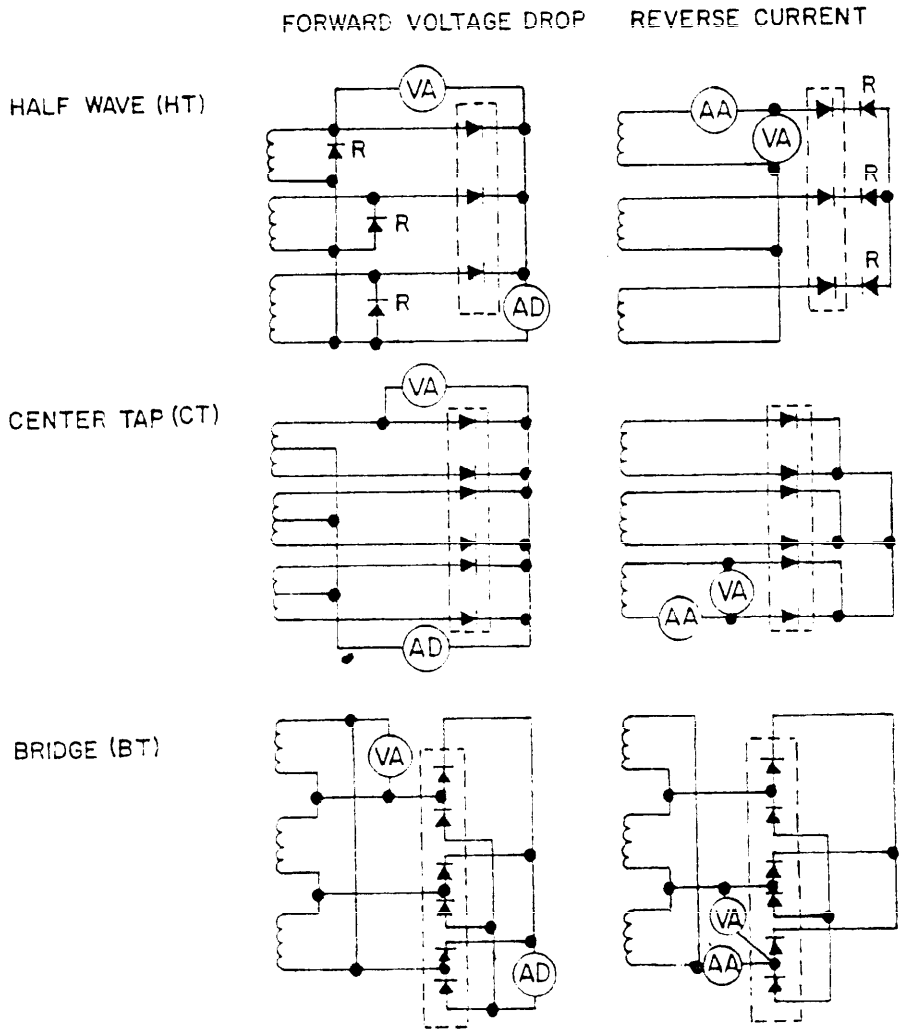
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AD – dc ammeter, permanent magnet, moving-coil type
AA – ac ammeter or milliammeter, rms reading
VA – ac voltmeter, VTVM
R – load balance

FIGURE 2. Test circuits for forward voltage drop and reverse current, single phase.

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AD – dc ammeter, permanent magnet, moving-coil type
AA – ac ammeter or milliammeter, rms reading
VA – ac voltmeter, VTVM
R – load balance

FIGURE 3. Test circuits for forward voltage drop and reverse current, three phase.

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TABLE IX. Dielectric test voltages.

AC applied voltage rating	Test potential
Volts rms	Volts rms
60 and under - - - - - - - - -	600
Over 60 to 90 inclusive - - -	900
Over 90 - - - - - - - - - - - -	1, 000 plus twice ac applied voltage rating, but shall not exceed 3, 000 volts

4.6.5 Insulation resistance. The insulation resistance between all current-carrying parts connected together and the mounting members shall be measured, using a 500-volt insulation tester (see 3.8).

4.6.6 Low-temperature exposure. Rectifiers shall be exposed to an ambient temperature of $-65^{\circ}\ ^{+0}_{-3}{}^{\circ}$C for 2 hours. At the conclusion of this conditioning, the rectifiers shall be removed from the conditioning chamber and allowed to remain at room ambient temperature for at least 4 hours. The rectifiers shall then be visually inspected, after which the forward voltage drop shall be measured as specified in 4.6.2 (see 3.9).

4.6.7 Moisture resistance. Rectifiers shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - On racks by normal mounting means.
(b) Initial measurements - Not applicable.
(c) Polarization and load - Not applicable.
(d) Final measurements - Rectifiers shall be tested for forward voltage drop, reverse current, dielectric strength, and insulation resistance as specified in 4.6.2, 4.6.3, 4.6.4, and 4.6.5, respectively (see 3.10).

4.6.8 Low-temperature operation. Rectifiers shall be exposed to an ambient temperature of $-55^{\circ}\ ^{+0}_{-3}{}^{\circ}$C for 1 hour. The rectifiers shall then be operated at rated current and voltage (see 3.1) at an ambient temperature of $-55^{\circ}\ ^{+0}_{-3}{}^{\circ}$ for 2 hours and, while at this ambient temperature, the forward voltage drop and reverse current shall be measured as specified in 4.6.2 and 4.6.3, respectively (see 3.11).

4.6.9 High-temperature operation. Rectifiers shall be operated at an ambient temperature of $-70^{\circ}\ ^{+0}_{-3}{}^{\circ}$ for 4 hours. The input voltage shall be 80 percent of the specified ac applied voltage (see 3.1). The output current into a resistive load shall be adjusted to 50 percent of rated current (see 3.1). Following this conditioning, and while at this elevated temperature, the reverse current shall be measured as specified in 4.6.3 (see 3.12).

4.6.10 Life. Rectifiers shall be operated continuously for 3, 000 hours at rated current and voltage (see 3.1) into a resistive load, while maintained at an ambient temperature of $-55^{\circ}\ ^{+0}_{-3}{}^{\circ}$ C. The forward voltage drop shall be measured as specified in 4.6.2 during the first hour of operation, at intervals of 250 hours, and at the completion of the test (see 3.13).

4.6.11 Mechanical shock. Rectifiers shall be mounted in each of three mutually perpendicular planes on the mounting adapter shown on figure 6E of MIL-S-901 and shall be subjected to the type C test of MIL-S-901. The rectifiers shall then be tested for forward voltage drop, reverse current, and dielectric strength as specified in 4.6.2, 4.6.3, and 4.6.4, respectively (see 3.14).

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4.6.12 Vibration. Rectifiers shall be tested in accordance with method 201 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - By normal mounting means.
(b) Initial measurements - Not applicable.
(c) Motion - One hour in each of three mutually perpendicular directions (total of 3 hours).
(d) Final measurements - Rectifiers shall be visually examined for loosening of parts or mechanical damage, and tested for forward voltage drop, reverse current, and dielectric strength as specified in 4.6.2, 4.6.3, and 4.6.4, respectively (see 3.15).

4.6.13 Corrosion. Rectifiers shall be tested in accordance with method 101, test condition B, of MIL-STD-202. Rectifiers shall then be visually inspected, and tested for forward voltage drop, reverse current, and dielectric strength as specified in 4.6.2, 4.6.3, and 4.6.4, respectively (see 3.16).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Rectifiers shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Rectifiers shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packaging. Rectifiers shall be individually packaged in accordance with MIL-P-116, submethod IC-2, insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. Mounting brackets, screws, nuts, and other hardware shall be physically protected and enclosed within the unit containers in a manner that will not damage the rectifiers.

5.1.1.5 Intermediate packaging. Rectifiers, packaged as described in 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size, shape and quantities, shall be of minimum tare and cube and shall contain multiples of five unit packages, not to exceed 50 packages. No intermediate packaging is required when the total quantity shipped to a single destination is less than 50 unit packages.

5.1.2 Level C. Cleaned and dried rectifiers shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The packaged rectifiers shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closures and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joints with tape, two inches minimum width, conforming to PPP-T60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using non-metallic or tape banding only.

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5.2.2 Level B. The packaged rectifiers shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto. For Army procurements, fiberboard containers shall be class weather resistant, as specified in level A.

5.2.3 Level C. The packaged rectifiers shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Rectifiers, packed as specified in 5.2.1, shall be placed on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Rectifiers, packed as specified in 5.2.2, shall be palletized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic. For Army procurement, caps shall be weather resistant as specified in level A.

5.2.4.3 Level C. Rectifiers, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Navy procurements. For Navy procurements the use of polystrene loose fill material (such as strips, strands and beads) is prohibited for packaging and packing applications.

6. NOTES

6.1 Intended use. Rectifiers covered by this specification are intended primarily for use in power rectification. For application in such items as magnetic amplifiers or as dc blocking components, additional tests other than those specified herein would be necessary to determine the acceptability of the rectifiers. The rectifiers specified herein are not for use in design after date of the specification. They are authorized for use in the design contracts effective prior to or on the date of this specification and to support existing military equipment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable military specification sheet, and the complete type designation (see 1.2.1 and 3.1).
(c) Whether preservation, packaging, packing, and marking are for domestic shipment for immediate use, domestic shipment involving limited or extended storage, or for overseas shipment. (See section 5.)

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6.2.1 Indirect shipments. The preservation and packaging, packing and marking specified in section 5 apply only to direct purchases by or direct shipments to the Government and are not intended to apply to contracts or orders between the supplier and prime contractor.

6.3 Definitions.

6.3.1 Rectifying element. A rectifying element is a circuit element which conducts current effectively in one direction only.

6.3.2 Rectifying junction. A rectifying junction is the region which exhibits the asymmetrical conductivity in a rectifier.

6.3.3 Rectifier cell. A rectifier cell is an elementary rectifying device having one positive electrode, one negative electrode, and one rectifying junction.

6.3.4 Rectifier. A rectifier is a single structure of one or more rectifier cells which utilizes the asymmetrical conductivity of the junction between dissimilar solid conducting materials to obtain a current flow 1/ effectively in one direction only.

6.3.5 Rectifier assembly. A rectifier assembly is a combination of one or more rectifiers, including all of the rectifying elements used in one rectifier circuit.

6.3.6 Forward direction. Forward direction is the direction of lesser resistance to current flow through a rectifier cell, i.e., from the negative electrode to the positive electrode.

6.3.7 Forward current. Forward current is the current that flows through a rectifier cell in the forward direction.

6.3.8 Forward voltage drop. Forward voltage drop is the voltage drop which results from the flow of current through a rectifier cell in the forward direction.

6.3.9 Reverse direction. Reverse direction is the direction of greater resistance to current flow through a rectifier cell, i.e., from the positive electrode to the negative electrode.

6.3.10 Reverse voltage. Reverse voltage is the voltage which is applied to a rectifier cell in the reverse direction.

6.3.11 Reverse current. Reverse current is the current which flows through a rectifier cell in the reverse direction.

6.3.12 Applied voltage rating (rms). The rms applied voltage rating of a rectifier cell is the maximum sinusoidal rms voltage which can be applied to the cell under specified conditions.

6.3.13 Basic rating. The basic rating of a rectifier is the dc output current and alternating-current (ac) applied voltage at which the rectifier is rated for continuous operation at specified load with convection cooling in a 55° C ambient.

6.3.14 Half-wave rectifier. A half-wave rectifier is a rectifier which changes alternating current into pulsating unidirectional current, utilizing only one-half inch of each cycle.

6.3.15 Full-wave rectifier. A full-wave rectifier is a rectifier which changes alternating current into pulsating unidirectional current, utilizing both halves of each cycle.

6.3.16 Basic bridge rectifier. A basic bridge rectifier is a single-phase, full-wave rectifier made up of four elements, each element consisting of a single rectifier cell.

1/ Current flow is used herein in the conventional sense.

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6.3.17 Ambient temperature. Ambient temperature is the temperature of the medium into which the heat of the rectifier is dissipated.

6.4 First article. Invitations for bid should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government and that bidders offering such products who wish to rely on such production or test must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.5 Current ratings. Current ratings of rectifiers covered by this specification are listed in table X.

TABLE X. Current ratings.

Style	Current ratings (amperes) 1/	
	Half wave	Bridge
RS10	0.100	0.200
RS20	.200	.400
RS30	.400	.800
RS40	.600	1.200
RS50	1.250	2.500
RS60	2.500	5.000
RS70	3.750	7.500
RS80	4.750	9.500

1/ The current values are for rectifiers having not more than 12 cells, operated continuously at an ambient temperature of $55^{\circ} {}^{+0}_{-3}{}^{\circ}$C.

Custodians:
Army - EL
Navy - EC

Review activities:
Army - MI, AT, AV, EL
Navy - EC, OS
DSA - ES

User activities:
Army - MU
Navy - SH, AS, CG, MC

Preparing activity:
Army - EL

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Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-R-11050B RECTIFIERS, METALLIC, SELENIUM

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

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